Primer: Nested Fixed Point (NFXP-NK)

Rust (1987), Iskhakov, Rust and Schjerning (2016)

econirl package documentation

1 Context and Problem

Harold Zurcher manages a fleet of buses and must decide each period whether to keep operating the current engine or pay a fixed cost to replace it. **?** formalized this as a dynamic discrete choice (DDC) model and proposed the first computationally feasible estimator: the nested fixed point algorithm. The outer loop maximizes the log-likelihood over structural parameters θ. The inner loop solves the Bellman equation to machine precision at each candidate θ, producing the choice probabilities that enter the likelihood.

The inner loop is the bottleneck. Value iteration converges at rate β per step, so when β approaches 1 (patient agents, long horizons), convergence slows dramatically. At $\beta = 0.9999$, successive approximation requires thousands of contractions per likelihood evaluation.

? introduced the SA-then-NK polyalgorithm that eliminates this bottleneck. The algorithm starts with successive approximation (SA) for safe global convergence and switches to Newton-Kantorovich (NK) near the fixed point for quadratic convergence. The total iteration count becomes insensitive to β: roughly 20 SA steps followed by 4 NK iterations suffices at any discount factor. In Monte Carlo experiments at $\beta = 0.9999$, NFXP-NK converges in 100 percent of 1,250 trials in 0.074 seconds, while MPEC-AMPL converges in zero.

NFXP is the foundational structural estimator in DDC. Every other estimator in this package defines itself relative to NFXP: CCP avoids the inner loop via Hotz-Miller inversion, MPEC replaces it with constrained optimization, NNES approximates it with a neural network, and TD-CCP eliminates the need for transition probabilities entirely.

2 Model and Notation

Table 1: Common DDC notation shared across all primers.

Symbol	Name	Definition
$s \in \mathcal{S}$	State	Discrete or continuous
$a \in \mathcal{A}$	Action	Finite: $\lvert\mathcal{A}\rvert = J$
s'	Next state	Drawn from $P(s' \mid s, a)$
$\beta \in [0, 1)$	Discount factor	
$\sigma > 0$	Scale parameter	Logit noise scale (= 1 by convention)
$\theta \in \Theta \subset \mathbb{R}^{d_\theta}$	Structural parameters	Unknown
$\phi(s, a)$	Feature vector	Known functions of (s, a), dimension K
$u(s, a; \theta)$	Flow utility	$\theta^\top \phi(s, a)$ (linear-in-parameters)
$P(s' \mid s, a)$	Transition kernel	Known or estimated from data
$Q(s, a)$	Choice-specific value	$u(s, a; \theta) + \beta \sum_{s'} P(s' \mid s, a)\, V(s')$
$V(s)$	Integrated value	$\sigma \log \sum_a \exp(Q(s, a)/\sigma)$
$\pi(a \mid s)$	Choice probability	$\exp\big((Q(s, a) - V(s))/\sigma\big)$
τ	Trajectory	$(s_0, a_0, s_1, a_1, \ldots)$ from one agent
$\mathcal{D}$	Panel data	Collection of N observed trajectories

An agent observes state s_t, chooses action a_t, and receives flow payoff $u(s_t, a_t; \theta) + \sigma\, \varepsilon_{t,a}$ where $\varepsilon_{t,a}$ is i.i.d. Type I extreme value. The choice-specific value function satisfies the Bellman equation

$$Q(s, a; \theta) \;=\; u(s, a; \theta) \;+\; \beta \sum_{s'} P(s' \mid s, a)\, V(s'), \tag{1}$$

where the integrated value function is the log-sum-exp

$$V(s) \;=\; \sigma \log \sum_a \exp\bigl(Q(s,a;\theta)/\sigma\bigr). \tag{2}$$

The optimal policy is the softmax $\pi(a \mid s;\theta) = \exp\bigl((Q(s,a) - V(s))/\sigma\bigr)$. Substituting (**??**) into (**??**) defines the soft Bellman operator T:

$$(TV)(s) \;=\; \sigma \log \sum_a \exp\Bigl(\bigl[u(s,a;\theta) + \beta \textstyle\sum_{s'} P(s'|s,a)\, V(s')\bigr]/\sigma\Bigr). \tag{3}$$

The operator T is a contraction with modulus β under the sup norm, so $V = TV$ has a unique fixed point for each θ.

3 Identification and Theory

Given panel data $\mathcal{D} = \{(s_i, a_i)\}_{i=1}^N$, NFXP estimates θ by maximizing the conditional log-likelihood

$$\hat{\theta}_{\text{NFXP}} \;=\; \arg\max_\theta \; \ell(\theta) \;=\; \arg\max_\theta \sum_{i=1}^N \log \pi(a_i \mid s_i;\theta). \tag{4}$$

Proposition 1 (MLE consistency, **?**)**.** *Under additive separability, conditional independence, and i.i.d. Type I extreme value errors, if the feature matrix $\Phi \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|\times K}$ has full column rank and Θ is compact with θ_0 interior, then $\hat{\theta}_{NFXP}$ is consistent and asymptotically efficient: $\sqrt{N}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1})$ where $I(\theta_0) = \mathbb{E}[\psi_i \psi_i^\top]$ is the Fisher information and ψ_i is the score* (**??**)*.*

The rank condition requires features that vary across actions. The discount factor β is not identified without restrictions (**?**) and must be fixed by the researcher. The scale σ is normalized to 1.

For counterfactual analysis, the identified object $\hat{\theta}$ supports policy simulation under parameter perturbation. Changing θ while holding P fixed produces a new policy by re-solving the Bellman equation. The parametric form $u = \theta^\top \phi$ resolves reward shaping ambiguity: the reward is pinned down by θ and ϕ, so there is no free shaping potential.

4 Estimation Algorithm

The gradient is computed analytically via the implicit function theorem, differentiating through the Bellman fixed point. The per-observation score is

$$\psi_i(\theta) \;=\; \frac{1}{\sigma}\Bigl[\nabla_\theta Q(s_i,a_i) - \sum_a \pi(a \mid s_i)\, \nabla_\theta Q(s_i,a)\Bigr], \tag{5}$$

where $\nabla_\theta Q(s,a) = \phi(s,a) + \beta \sum_{s'} P(s'|s,a) \frac{\partial V}{\partial \theta}$ and the total derivative solves the linear system

$$(I - \beta P_\pi)\frac{\partial V}{\partial \theta} = \sum_a \pi(a|s)\, \phi(s,a). \tag{6}$$

The BHHH outer optimizer approximates the Hessian with the outer product of scores $B_k = \frac{1}{N}\sum_i \psi_i \psi_i^\top$, which is guaranteed positive definite.

Algorithm 1: NFXP-NK (**??**)

Input: Panel $\mathcal{D} = \{(s_i, a_i)\}_{i=1}^N$, features ϕ, transitions P, discount β, scale σ
Initialize θ_0 from data heuristic
for $k = 0, 1, 2, \ldots$ **do**
  Inner loop: solve $V = TV$ at θ_k
  begin `// SA phase: safe global convergence`
    for $j = 1, 2, \ldots$ **to** *convergence* **do** $V \leftarrow TV$
  end
  begin `// NK phase: quadratic local convergence`
    for $j = 1, 2, \ldots$ **do**
      Solve $(I - \beta \nabla T)\delta = TV - V; \quad V \leftarrow V + \delta$
      if $\|\delta\|_\infty < \varepsilon_{inner}$ **then break**
    end
  end
  Compute $Q(s, a)$ via (**??**), $\pi(a|s)$ via softmax
  Score: solve (**??**) for $\partial V / \partial \theta$, compute ψ_i via (**??**)
  Outer update (BHHH):
  $B_k \leftarrow \frac{1}{N} \sum_i \psi_i \psi_i^\top + \lambda I$
  $d_k \leftarrow B_k^{-1} (\frac{1}{N} \sum_i \psi_i)$
  Line search: α_k via step-halving until $\ell(\theta_k + \alpha_k d_k) > \ell(\theta_k)$
  $\theta_{k+1} \leftarrow \theta_k + \alpha_k d_k$
  if $\|\nabla \ell\| < \varepsilon_{outer}$ **then break**
end
Output: $\hat{\theta}$, V^*, π^*, B^{-1} (robust standard errors)

Default tolerances: $\varepsilon_{\text{switch}} = 10^{-3}$, $\varepsilon_{\text{inner}} = 10^{-12}$, $\varepsilon_{\text{outer}} = 10^{-6}$, $\lambda = 10^{-8}$. Code: `src/econirl/estimation/nfxp.py`, inner loop in `src/econirl/core/solvers.py`.

5 Simulation

Table 2: Rust bus engine (90 states, $\beta = 0.95$, 50,000 obs). True parameters: $\theta_c = 0.001$, $RC = 3.0$.

Metric	BC	NFXP-NK
$\hat{\theta}_c$	—	0.001044 (0.000253)
$\widehat{RC}$	—	3.0079 (0.0312)
Log-likelihood	−10130.62	−10156.55
Policy accuracy	98.9%	100.0%
Analytical SEs	No	Yes
Time (s)	0.50	2.56

NFXP recovers both parameters within one standard error of the true values. The operating cost estimate is 0.001044 (true 0.001) and the replacement cost is 3.0079 (true 3.0), both with analytical standard errors from the BHHH outer product. Policy accuracy is 100.0%: the estimated policy assigns the same greedy action as the true policy at every state. Behavioral cloning matches the greedy policy at 98.9% of states but cannot recover structural parameters or produce standard errors, confirming that NFXP is learning from MDP structure.

NFXP's advantages over CCP and MPEC are threefold: the inner loop produces an exact fixed point so the likelihood surface is smooth, the analytical score (**??**) gives exact gradients at negligible cost, and the BHHH outer product produces valid robust standard errors without bootstrap. The limitation is computational scaling: the inner loop costs $O(|\mathcal{S}|^2)$ per contraction and the NK step costs $O(|\mathcal{S}|^3)$. For state spaces above a few thousand, NNES, TD-CCP, and SEES scale better.

Code. `nfxp_run.py` generates Table **??**. Estimator: `src/econirl/estimation/nfxp.py`.

```
.venv/bin/python papers/econirl_package/primers/nfxp/nfxp_run.py
```